

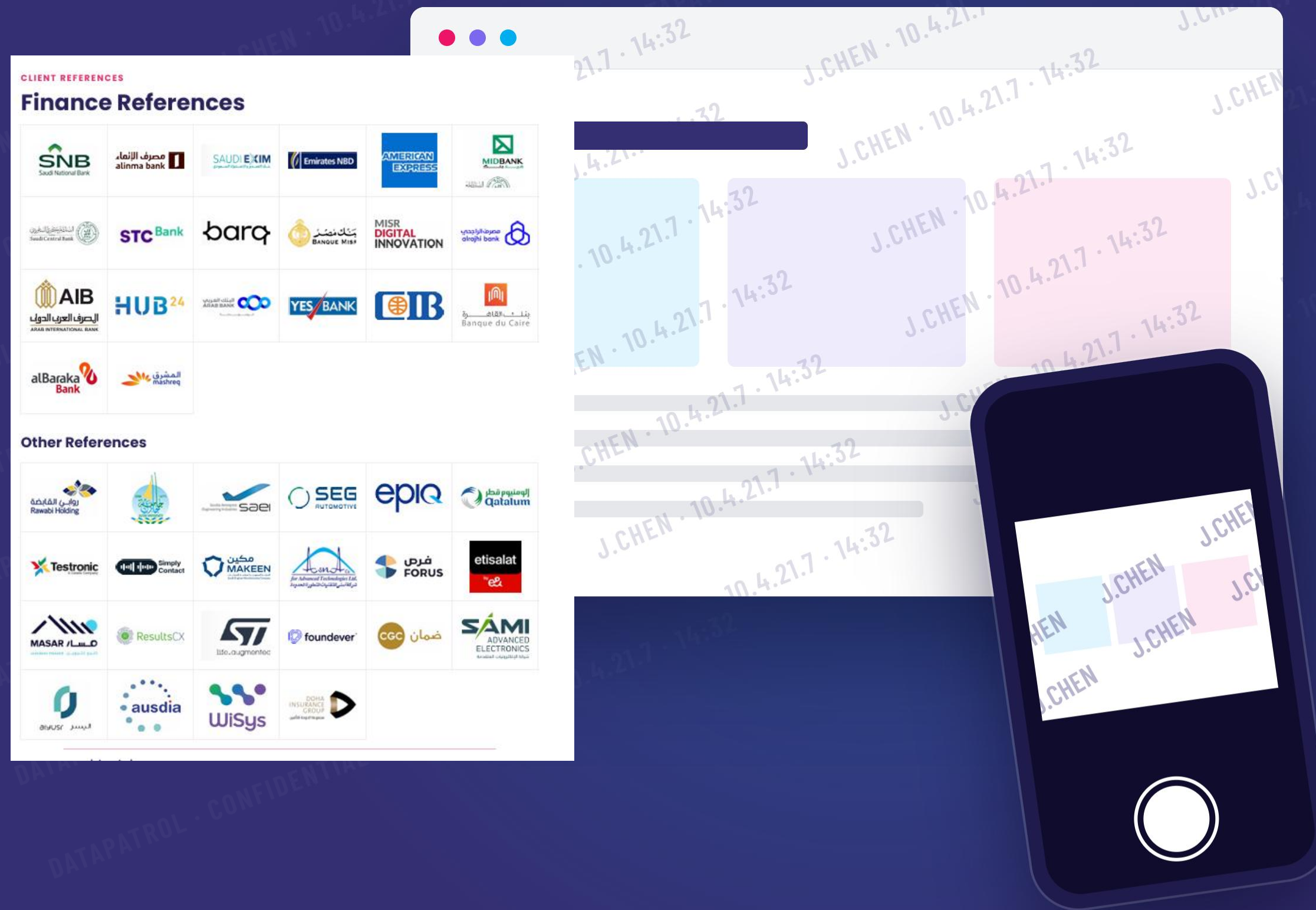
DATA PATROL

PROTECTING IN VIEW

Visual data security for the
insider-risk era

Presented by Omar Samhouri

www.datapatrol.com



A VISUAL LEAK LEAVES NO LOG



SECURITY EVENT LOG		
14:32:01	Customer export opened	LOGGED

A VISUAL LEAK LEAVES NO LOG



SECURITY EVENT LOG

14:32:01	Customer export opened	LOGGED
14:32:04	Phone raised to the screen	NO EVENT

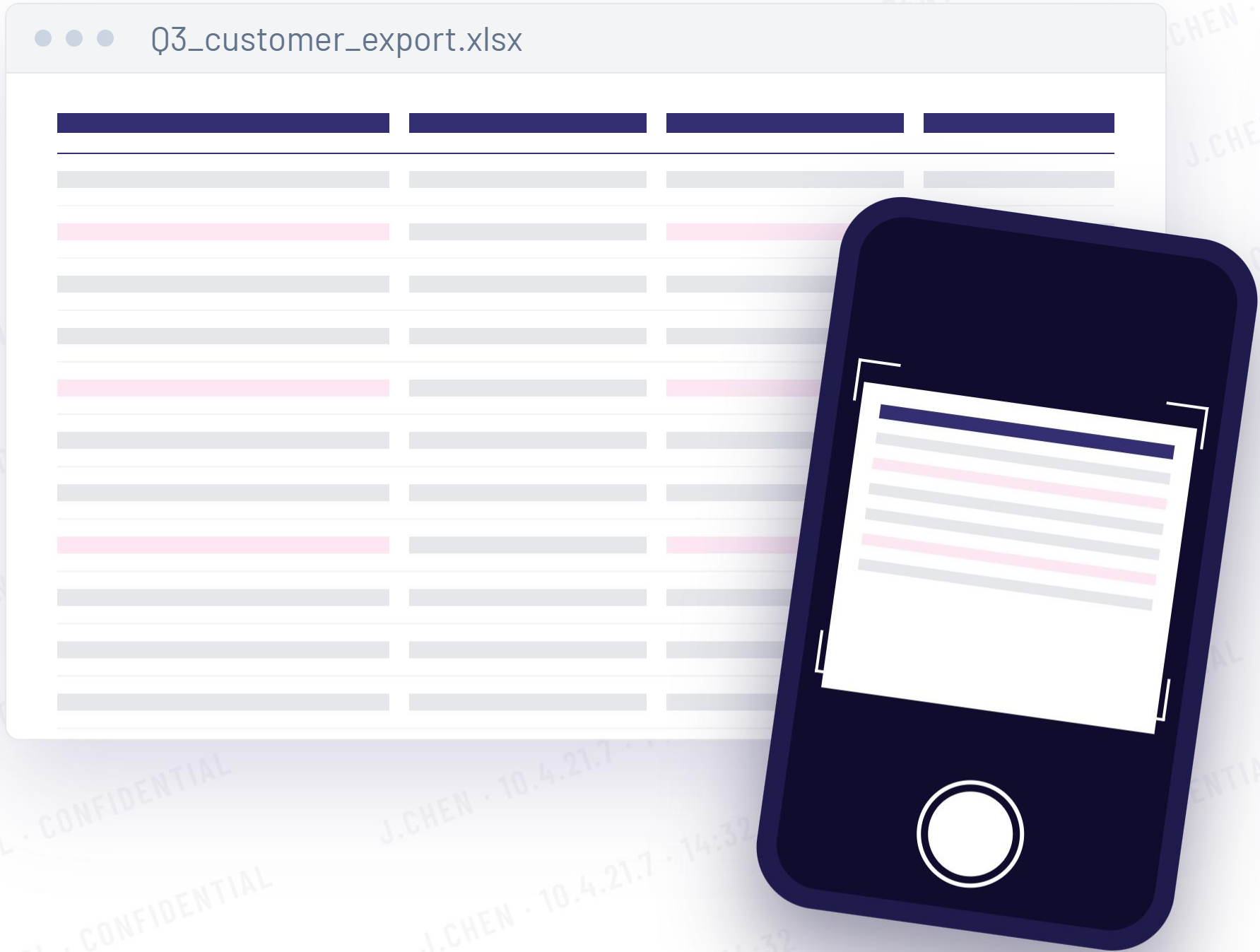
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SECURITY EVENT LOG

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14:32:04	Phone raised to the screen	NO EVENT
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14:32:11	Photo sent from a personal chat app	NO EVENT

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14:32:04	Phone raised to the screen	NO EVENT
14:32:05	Photo taken	NO EVENT
14:32:11	Photo sent from a personal chat app	NO EVENT

DLP, SIEM and endpoint tools saw one event: the file opening.

VISUAL LEAKS SUCCEED, AND MOSTLY GO UNNOTICED



91%

of visual-hacking attempts in a global field study succeeded

Ponemon Institute / 3M Visual Hacking Experiment, 2016 · Ponemon Institute, 2025

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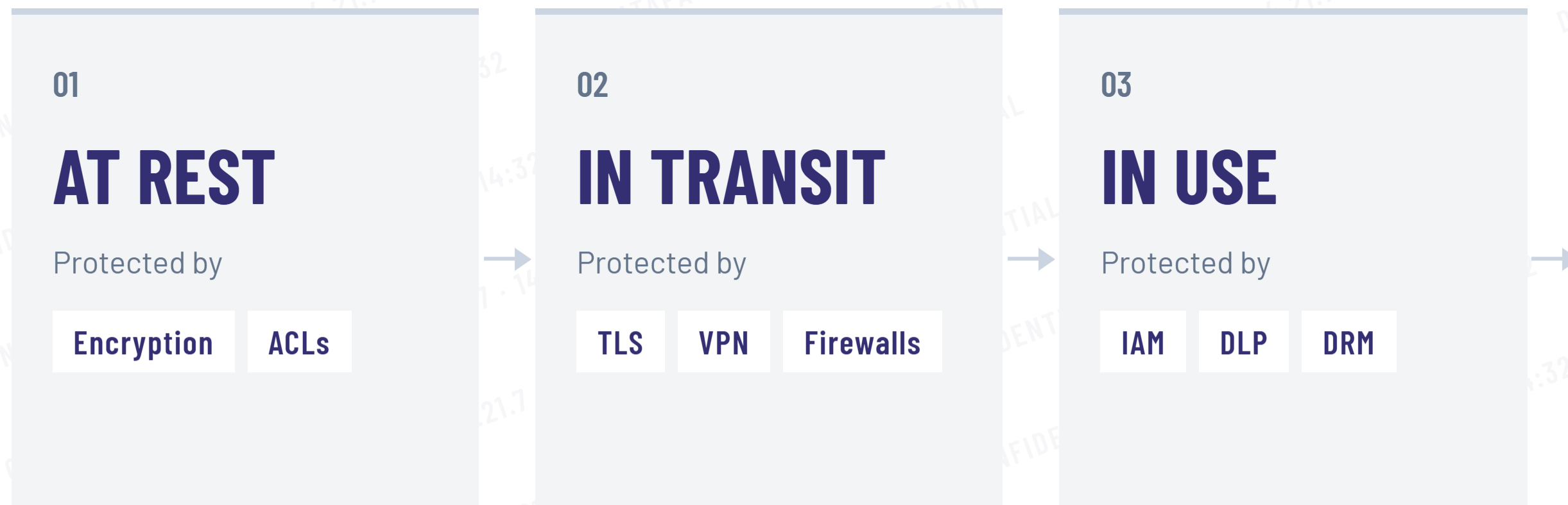
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\$17.4M

average annual cost of insider incidents

Ponemon Institute / 3M Visual Hacking Experiment, 2016 · Ponemon Institute, 2025

THE FOURTH STATE OF DATA **HAS NO PROTECTION**



Data in view is anything rendered on a monitor, projector, VDI session or shared screen: the moment it reaches human eyes and cameras.

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1,000 PHONE PHOTOS OF 2NM PROCESS DATA

MALICIOUS · TSMC, 2023-2025

HOW

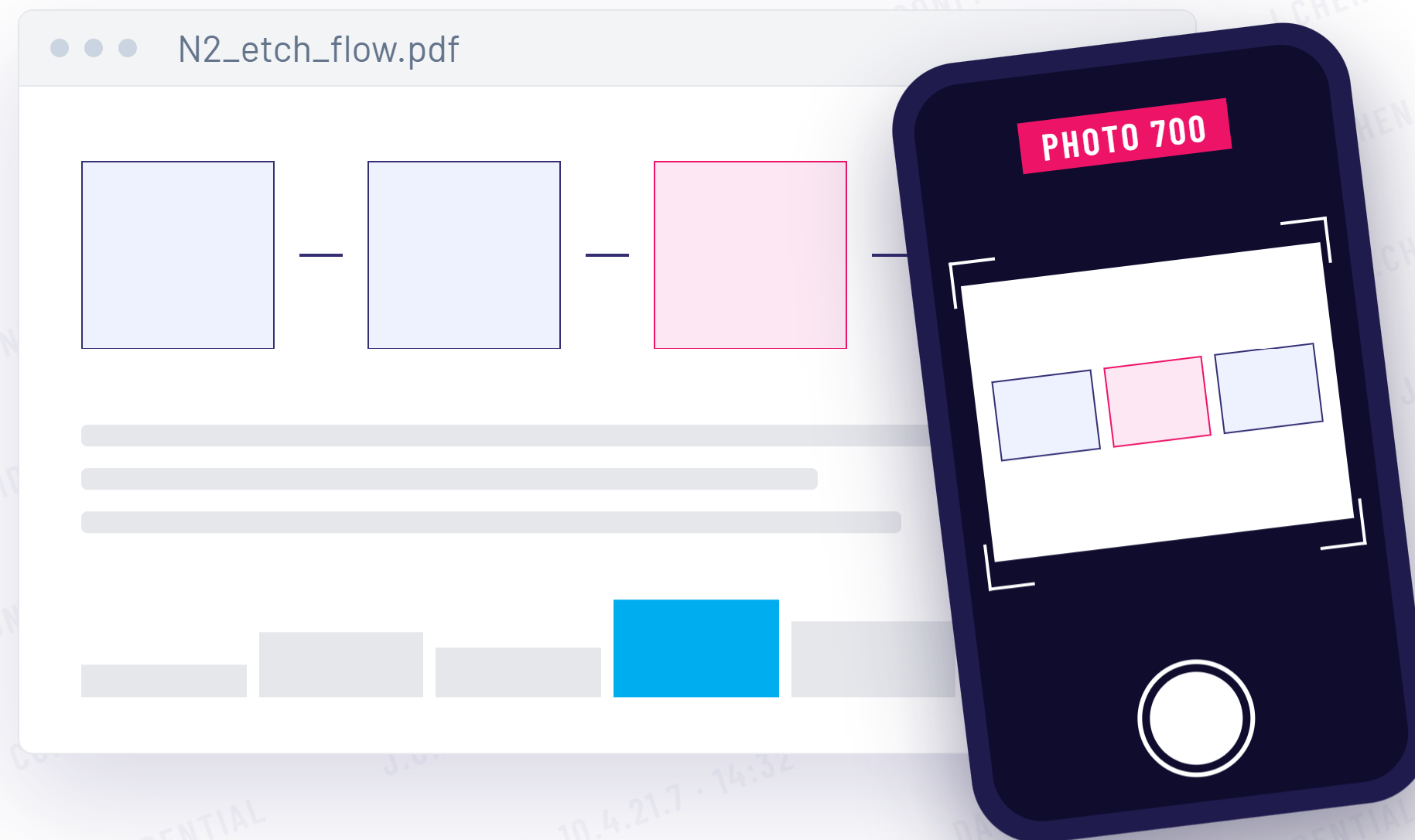
A former engineer asked current staff to open confidential process diagrams on screen, then photographed them with his phone

SCALE

About 1,000 images from two engineers

OUTCOME

Charged under national security law; ringleader sentenced to 10 years in April 2026



Sources: Taiwan News, Aug 2025 · Focus Taiwan, Apr 2026 · TechNode, Aug 2025

FIND THE LEAK

Meeting ID:

35 participants

 PM_office	 d.lindqvist	 m.okafor	 s.patel	 r.whitfield
 a.moreau	 cabinet.sec	 t.brennan	 l.cho	 k.osei
 e.ramsey	 h.vance	 g.ashdown	 n.fairweather	 j.harper

FIND THE LEAK

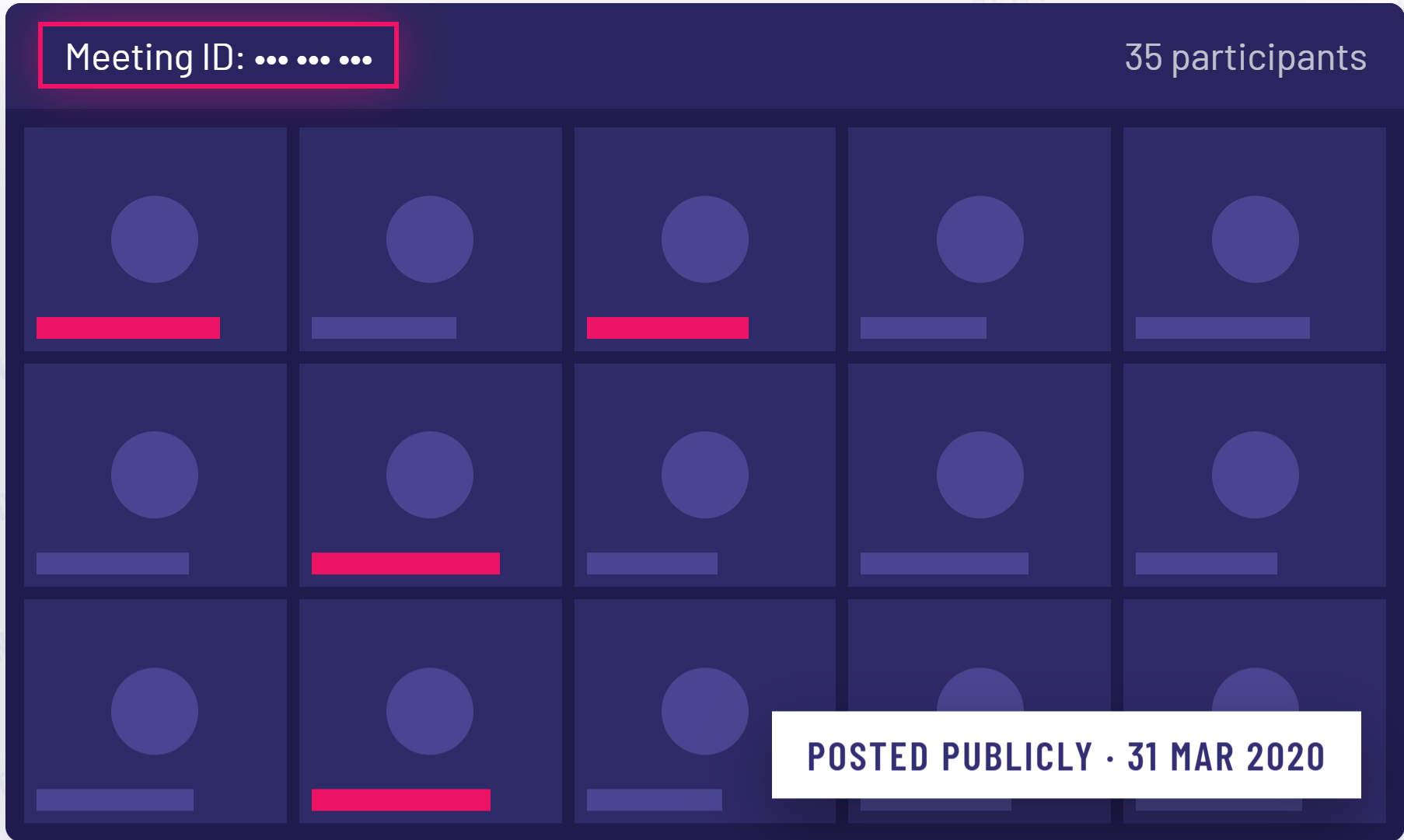
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e.ramsey	h.vance	g.ashdown	n.fairweather	j.harper	

VISIBLE TO EVERYONE

MEETING ID

MINISTERS'
USERNAMES

A CABINET MEETING ID IN A PUBLIC SCREENSHOT



ACCIDENTAL · UK CABINET, 2020

HOW

The Prime Minister shared a screenshot of the first digital Cabinet meeting on social media

EXPOSED

The Zoom meeting ID and ministers' usernames

OUTCOME

Security experts criticised the post within hours; Downing Street said the call was password protected

Sources: PA via Shropshire Star, 31 Mar 2020 · Graham Cluley, 31 Mar 2020

DIFFERENT INTENT, SAME EXIT POINT

CASE 1 · MALICIOUS

Intent	Deliberate theft
Channel	Phone camera
Needs	A reason not to try, and a trail if they do



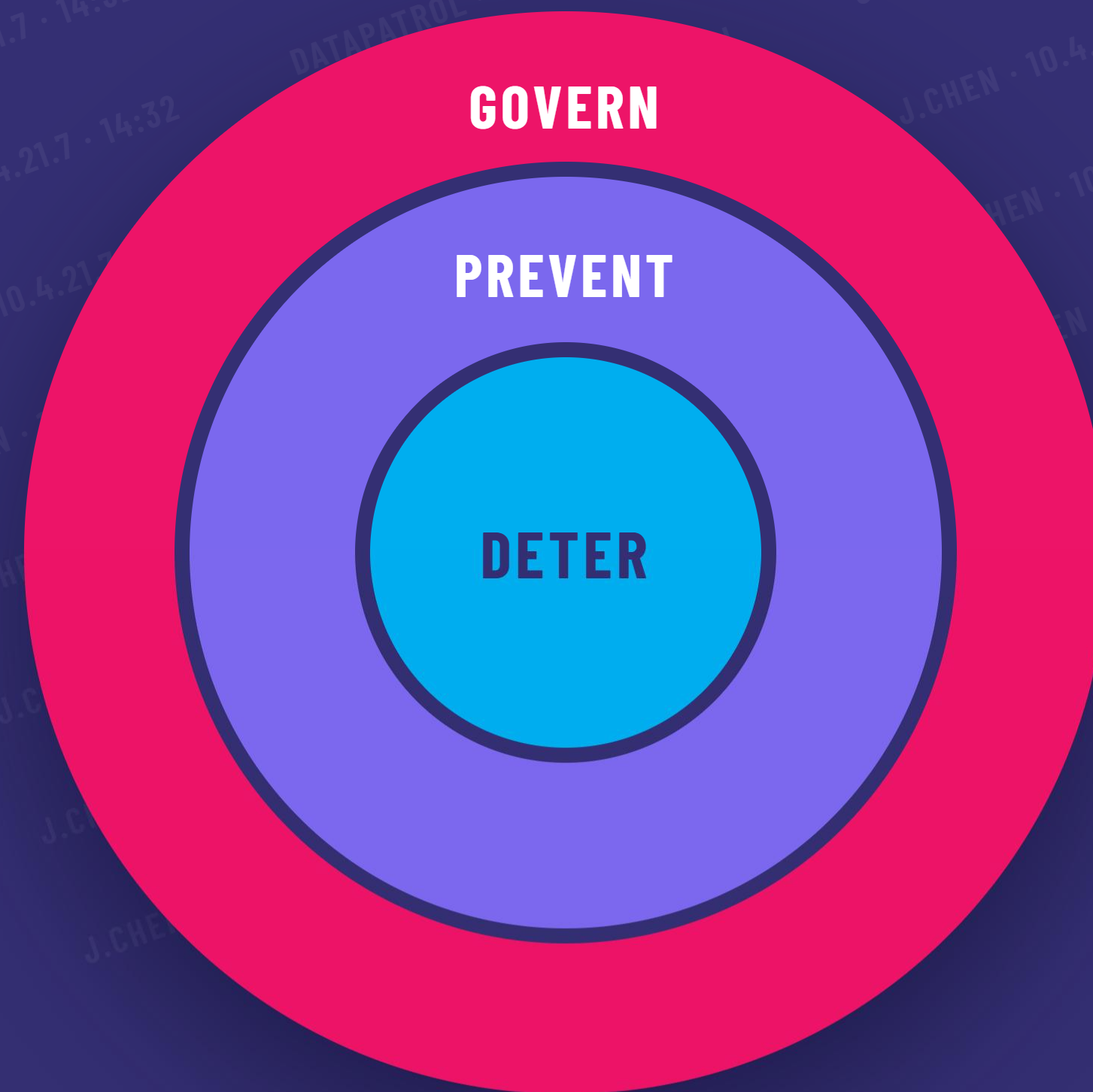
CASE 2 · ACCIDENTAL

Intent	None
Channel	Screenshot
Needs	A control that stops the capture before it happens

03

THREE LEVELS OF CONTROL

Deter. Deter and prevent. Or deter, prevent
and govern.



EACH LEVEL BUILDS ON THE ONE BEFORE

01

DETERRENCE

Make every leak traceable

WATERMARKING

On the screen or on the app window

EACH LEVEL BUILDS ON THE ONE BEFORE

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DETERRENCE

Make every leak traceable

WATERMARKING

On the screen or on the app window

02

PREVENTION

Stop sensitive data leaving the screen

Deterrence, plus

- + ANTICOPY
- + PRTSC PREVENTION
- + PROCESS PATROL

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03

GOVERNANCE

Control documents, policy and proof

Prevention, plus

- + PRINTING WATERMARK
- + POLICYPULSE
- + DOCUMARK

LIVE AUDIENCE MOMENT

**GO AHEAD.
TAKE A PHOTO
OF THIS SLIDE.**

10:41:16

LIVE ON STAGE · 1 Oct
2026

YOUR PHOTO JUST BECAME EVIDENCE



ZOOMED IN
LIVE ON STAGE
10:41:20

Event	Live on stage
Date	1 Oct 2026
Captured	10:41:20
Source	This room

Every photo taken a minute ago names this event and the moment it was taken. On a protected screen, the same mark names the viewer.

DETERRENCE MAKES EVERY LEAK TRACEABLE

A visible, tamper-resistant watermark puts the viewer's identity on everything they see. Anyone who photographs, records or shares the screen carries their name with it.

OPERATING MODES

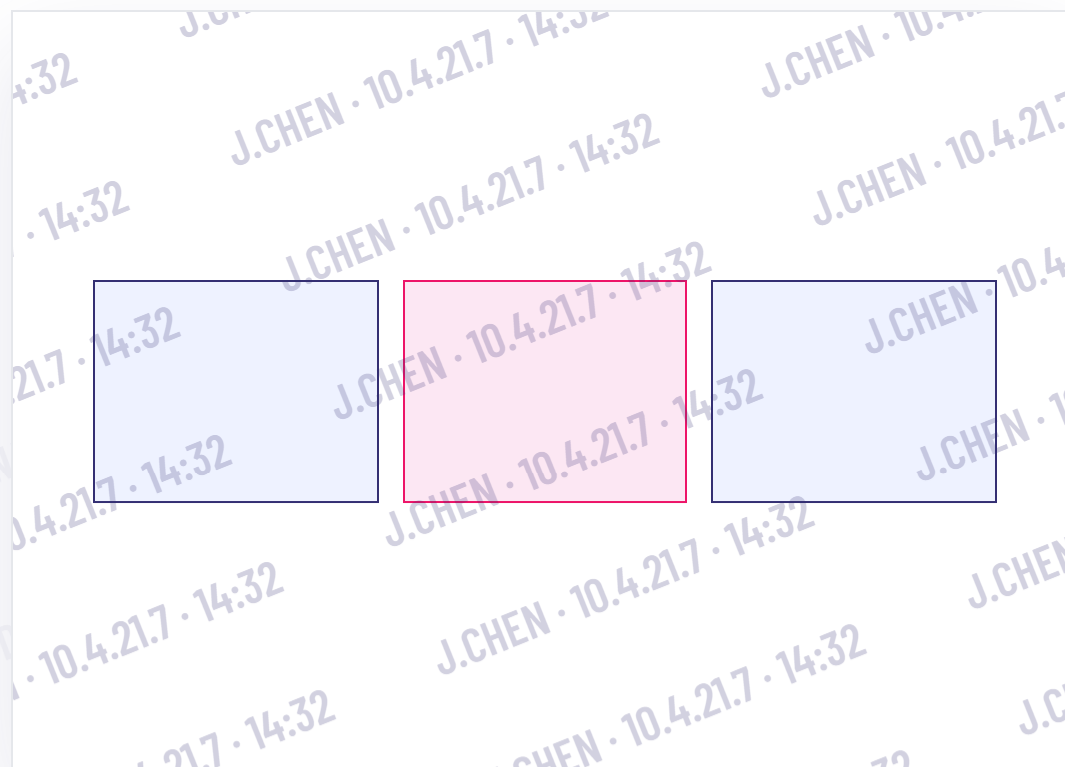
ON THE SCREEN

ON THE APP WINDOW

EVERY FRAME CARRIES

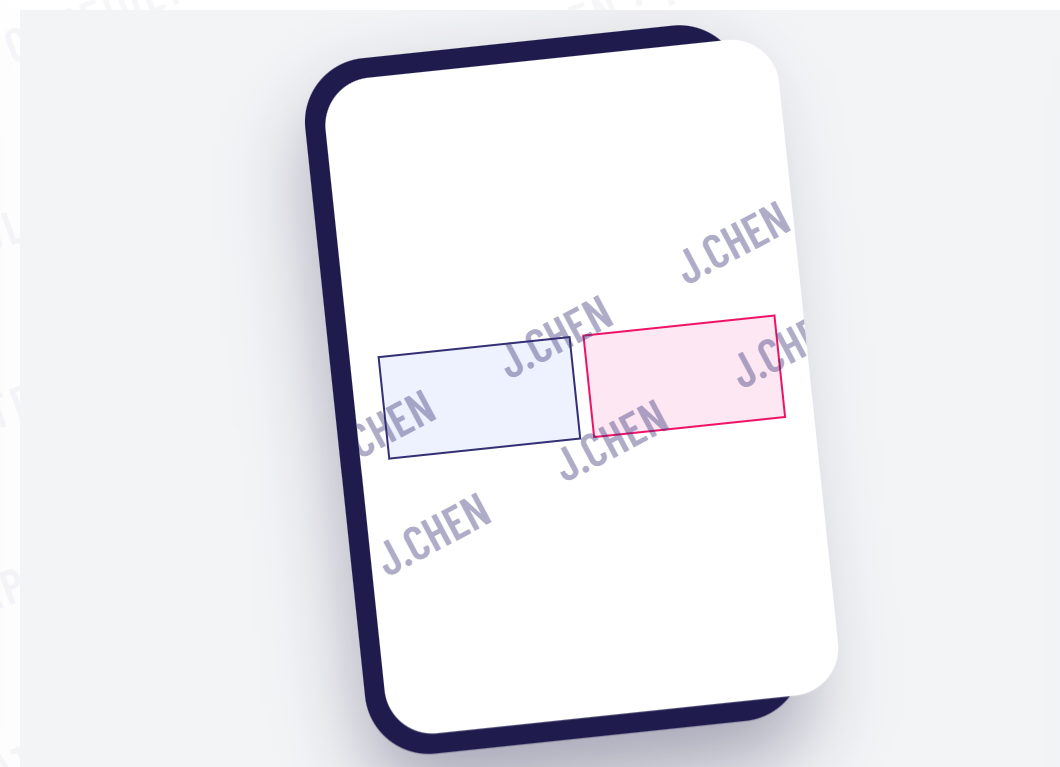
J.CHEN · 10.4.21.7 · 14:32

REPLAYING CASE 1 UNDER DETERRENCE



01 ON SCREEN

The diagram opens with the engineer's identity across it



02 PHOTOGRAPHED

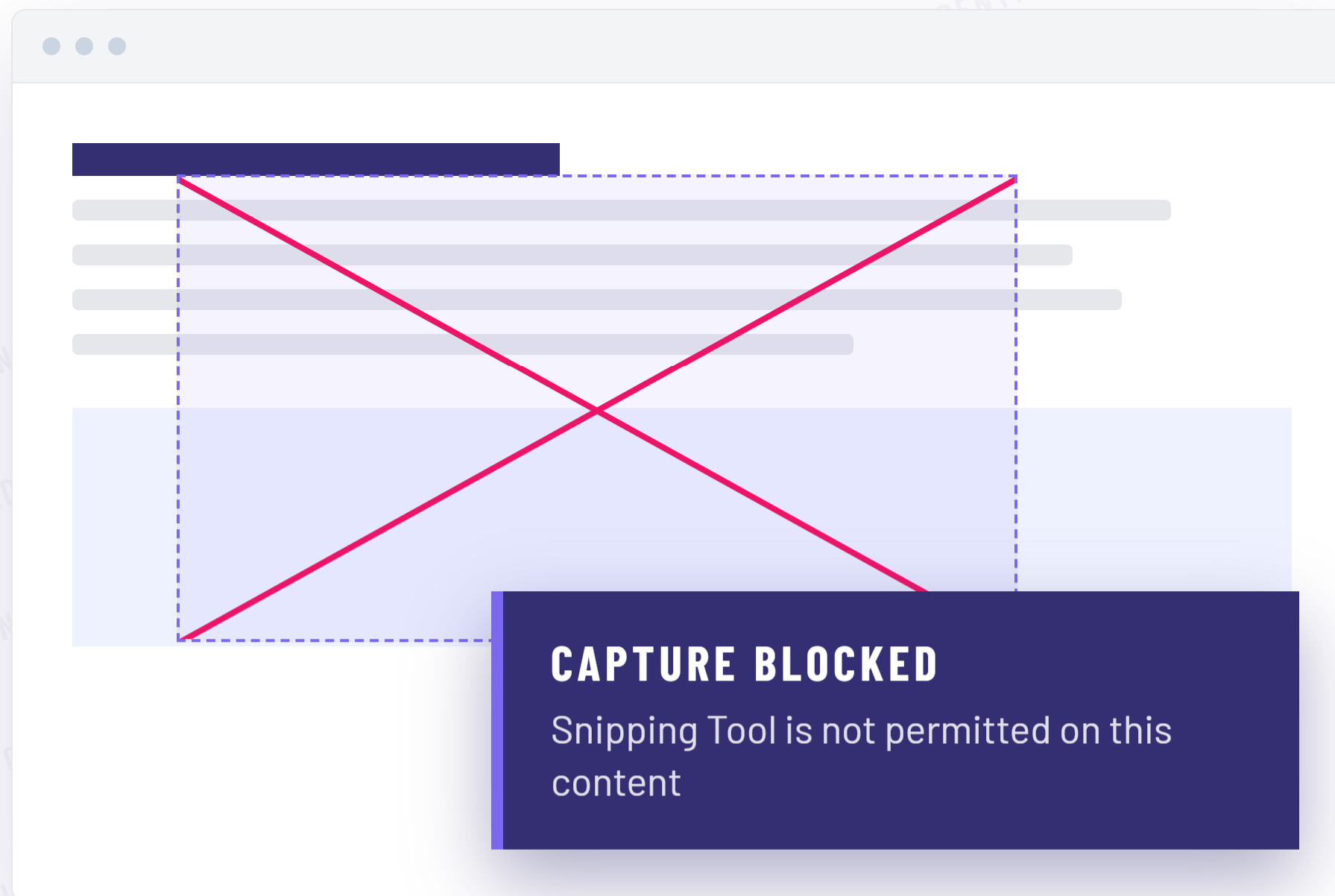
The watermark is in the photo, visible to the person taking it

User	J. Chen
Device	WS-2231
IP	10.4.21.7
Time	14:32:07

03 TRACED

Any image that surfaces points to its source user and moment

PREVENTION STOPS DATA LEAVING THE SCREEN



+ ANTICOPY

Blocks copy and cut of sensitive text by policy

+ PRTSC PREVENTION

Blocks PrintScreen, Snipping Tool and capture apps at the OS level

+ PROCESS PATROL

Kernel-level control over which apps can run

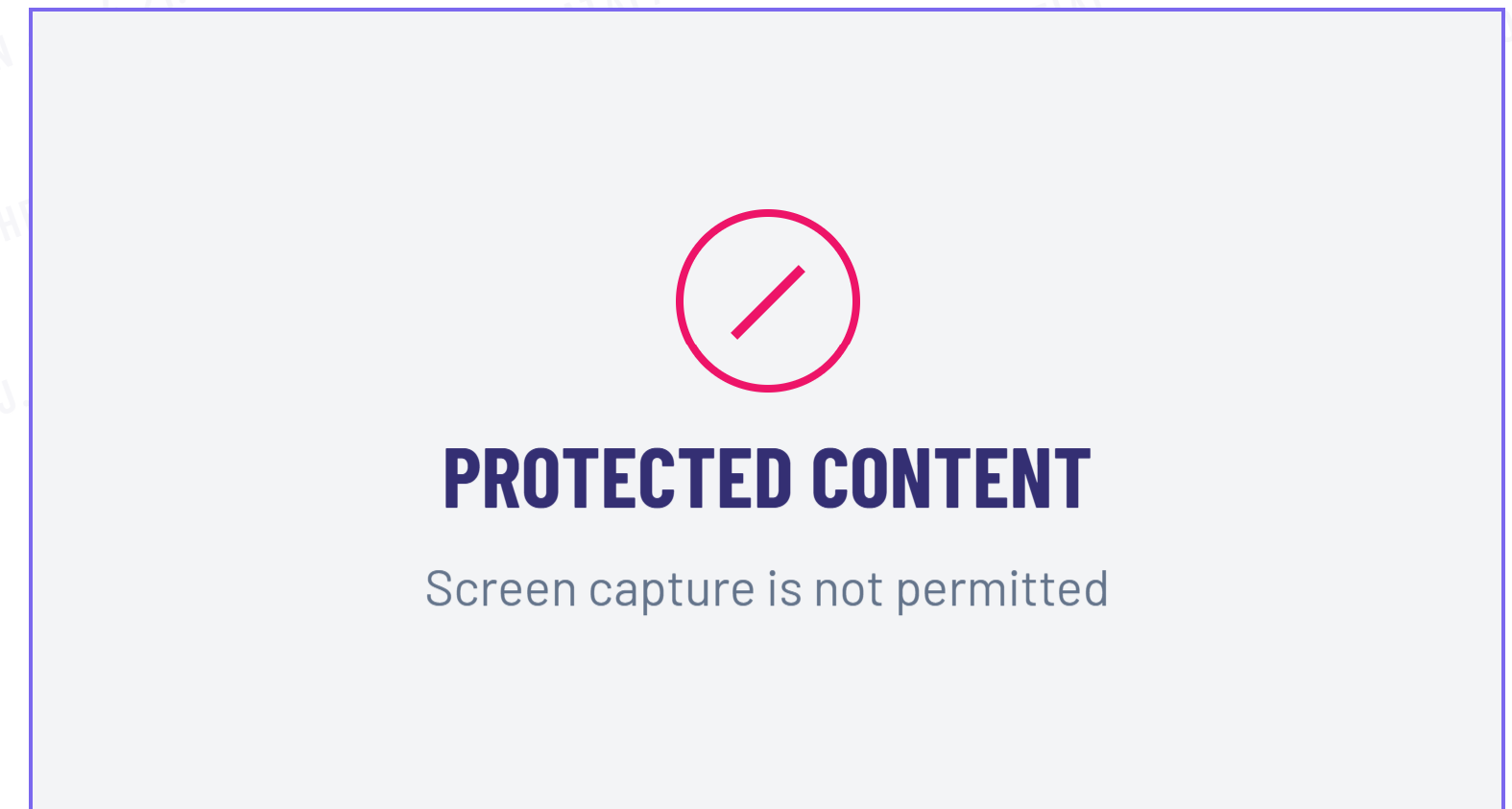
REPLAYING CASE 2 UNDER PREVENTION

WHAT WAS CAPTURED

Meeting ID:			

Meeting ID and usernames, posted publicly

WITH PREVENTION



The capture is blocked, so there is nothing to post

GOVERNANCE CONTROLS **POLICY AND PROOF**

+ **PRINTING WATERMARK**

Traceable marks on every print and PDF export

+ **DOCUMARK**

A watermark inside the file, wherever it is forwarded

+ **POLICYPULSE**

Signed, timestamped policy acknowledgment on screen



MATCHING THE LEVEL TO THE RISK

	01 DETER	02 PREVENT	03 GOVERN
Phone photo of the screen	■ Traced	■ Traced	■ Traced
Screen shared with outside guests	■ Traced	■ Traced	■ Traced
Screenshot or snipping tool	■ Traced	■ Blocked	■ Blocked
Copy-paste of sensitive text	—	■ Blocked	■ Blocked
Unapproved recording app	■ Traced	■ Blocked	■ Blocked
Printed page leaves the building	—	—	■ Traced
File forwarded outside	—	—	■ Traced
Proof that staff accepted policy	—	—	■ Signed

ADD-ONS CARRY ANY LEVEL TO WEB AND MOBILE



+ WEBMARK

A lightweight SDK that adds real-time watermarking to internal web apps and portals

Works with

DETER

PREVENT

GOVERN



+ MOBILEMARK

The same always-on watermark, carried into phone and tablet workspaces

Works with

DETER

PREVENT

GOVERN

AI CAN'T ERASE **THE EVIDENCE**



AI REMOVAL

STOPPED BY FULL COVERAGE

Inpainting can erase one instance. The mark repeats across the frame, with a pattern layer beneath the surface.

AI REGENERATION

FAILS VERIFICATION

A synthetic image never touched a protected screen, so it carries no genuine signature.

BUILT FOR REGULATED INDUSTRIES

01

FINANCIAL SERVICES

Call-center screens
Trading floor displays
Pre-earnings materials

02

HEALTHCARE

EHR access
Clinical trial data
Shared hospital terminals

03

GOVERNMENT

Classified and CUI workstations
Citizen data portals
Inter-agency screen shares

04

LEGAL

M&A data rooms
E-discovery vendor access
Sealed court filings

05

TECHNOLOGY

Source-code reviews
Product roadmaps
Customer data dashboards

DEPLOY ON-PREMISES OR IN THE CLOUD

ON-PREMISES

Installed in your own infrastructure

Data never leaves your network

Full control over updates and policy

SAAS / CLOUD

Hosted and managed by DataPatrol

Live in days, not months

Automatic updates and scaling

One agent, one console, either way. WebMark and MobileMark add-ons run on both models.

TRUSTED BY LEADING INSTITUTIONS

FINANCE

OTHER INDUSTRIES



WHICH LEVEL FITS YOUR RISK TODAY?

DETER

PREVENT

GOVERN

+ WEBMARK · MOBILEMARK

Your photo from 10:41:20 still shows exactly where it came from.